

Supplementary Information for

Resource competition between buoyancy-regulating and sinking phytoplankton species along a stratified water column

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S1. Influence of environmental parameters on biomass profiles over depth

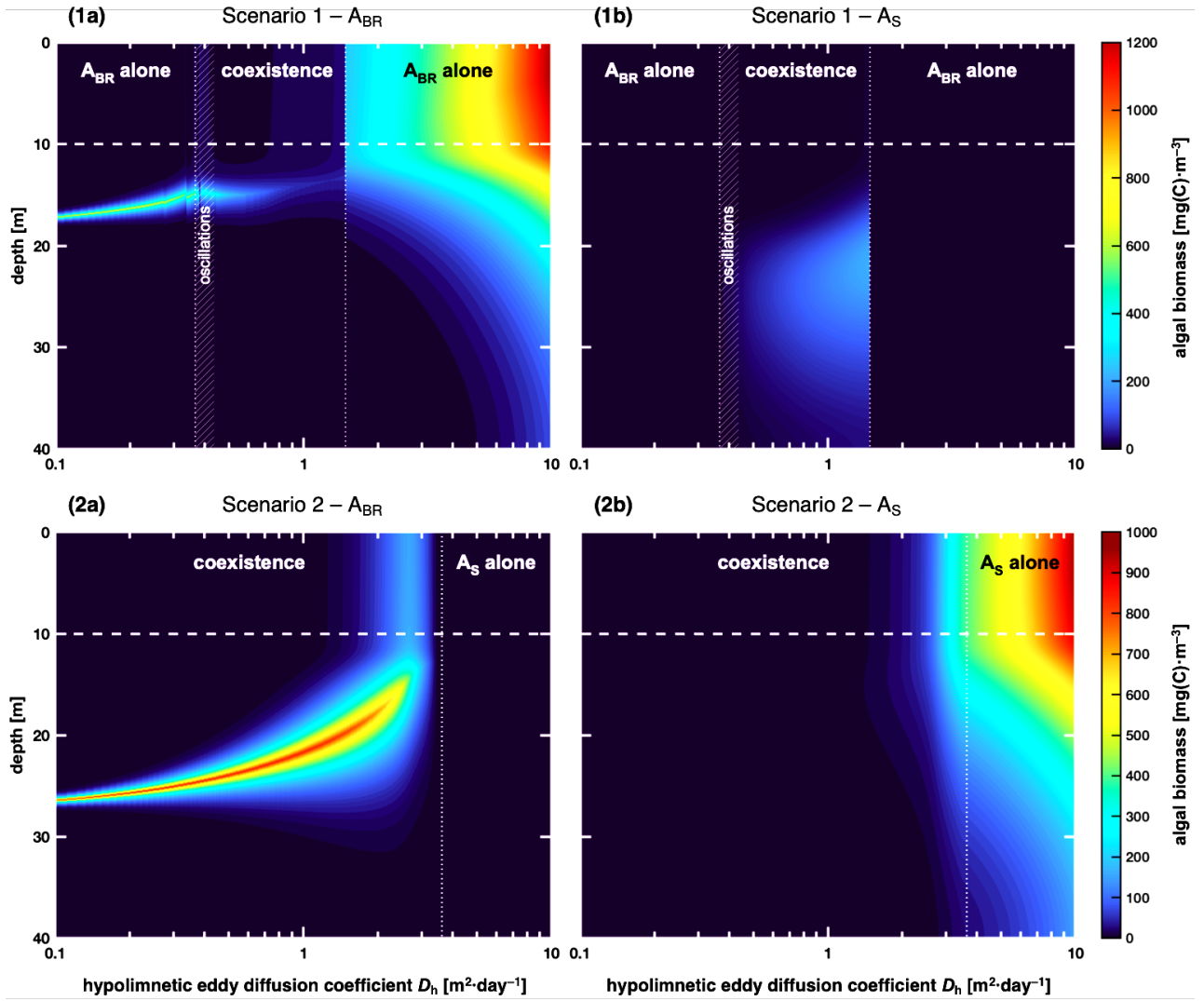


Figure S1. Influence of the environmental gradient of hypolimnetic eddy diffusion coefficient (D_h) on biomass profiles of (a) A_{BR} and (b) A_S for (1) Scenario 1 (upper row) and (2) Scenario 2 (lower row). In each contour plot, the vertical axis is the depth [m] from surface to the bottom of the water column, the horizontal axis is the gradient of D_h [$\text{m}^2 \cdot \text{day}^{-1}$], and the color scale indicates the local algal biomass concentration [$\text{mg(C)} \cdot \text{m}^{-3}$]. The horizontal dashed white line delineates the thermocline (separation between epilimnion and hypolimnion, fixed at 10 m), while vertical dotted white lines indicate the boundary between different system states. Parameters are taken from Tab. 1, with $a_{bg} = 0.2$ [m^{-1}] and $N_0 = 50$ [$\text{mg(P)} \cdot \text{m}^{-3}$].

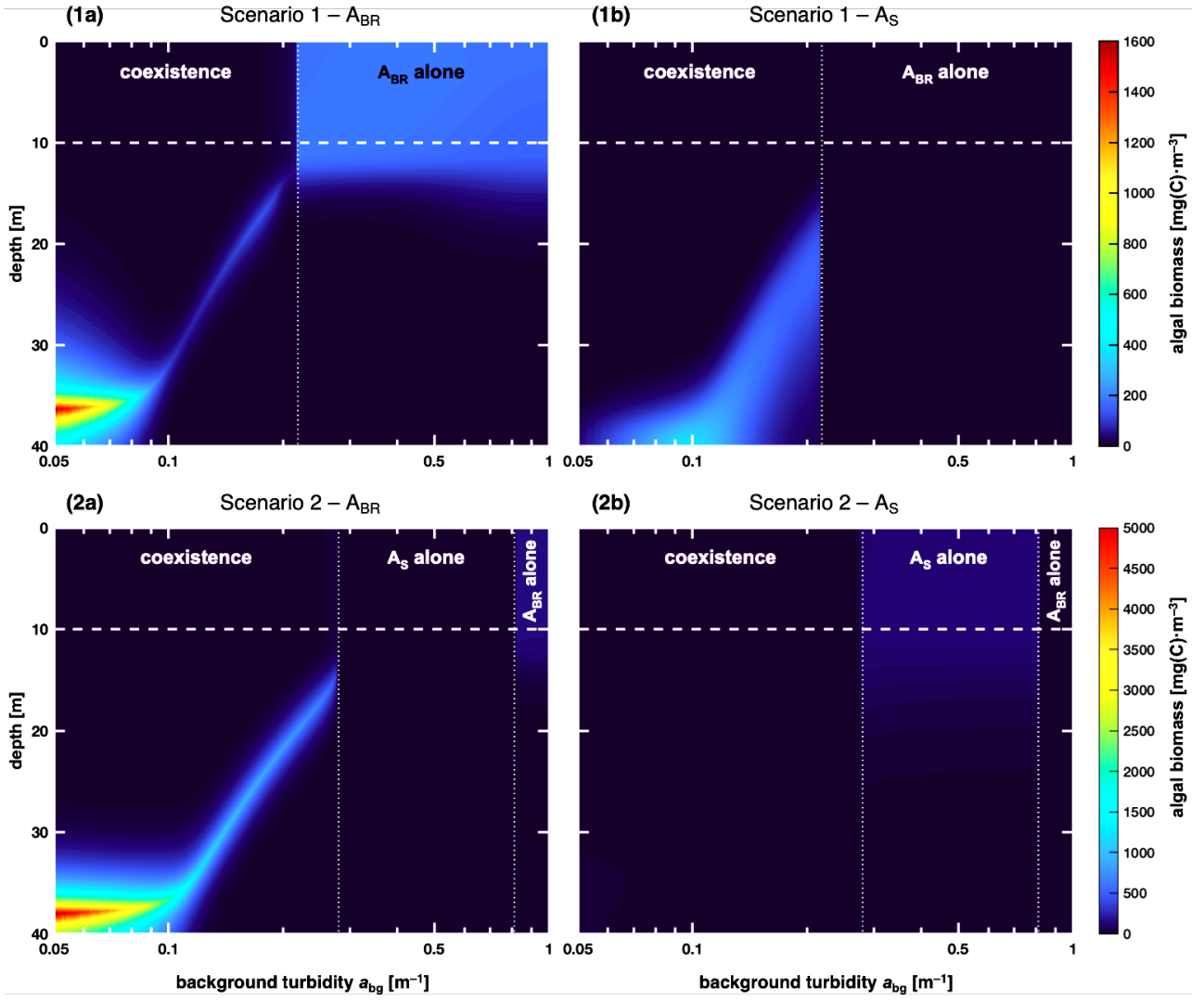


Figure S2. Influence of the environmental gradient of background turbidity (a_{bg}) on biomass profiles of **(a)** A_{BR} and **(b)** A_S for **(1)** Scenario 1 (upper row) and **(2)** Scenario 2 (lower row). In each contour plot, the vertical axis is the depth [m] from surface to the bottom of the water column, the horizontal axis is the gradient of a_{bg} [m^{-1}], and the color scale indicates the local algal biomass concentration [$mg(C) \cdot m^{-3}$]. The horizontal dashed white line delineates the thermocline (separation between epilimnion and hypolimnion, which is fixed at 10 m), while vertical dotted white lines indicate the boundary between different system states. Parameters are taken from Tab. 1, with $D_h = 1$ [$m^2 \cdot day^{-1}$] and $N_0 = 50$ [$mg(P) \cdot m^{-3}$].

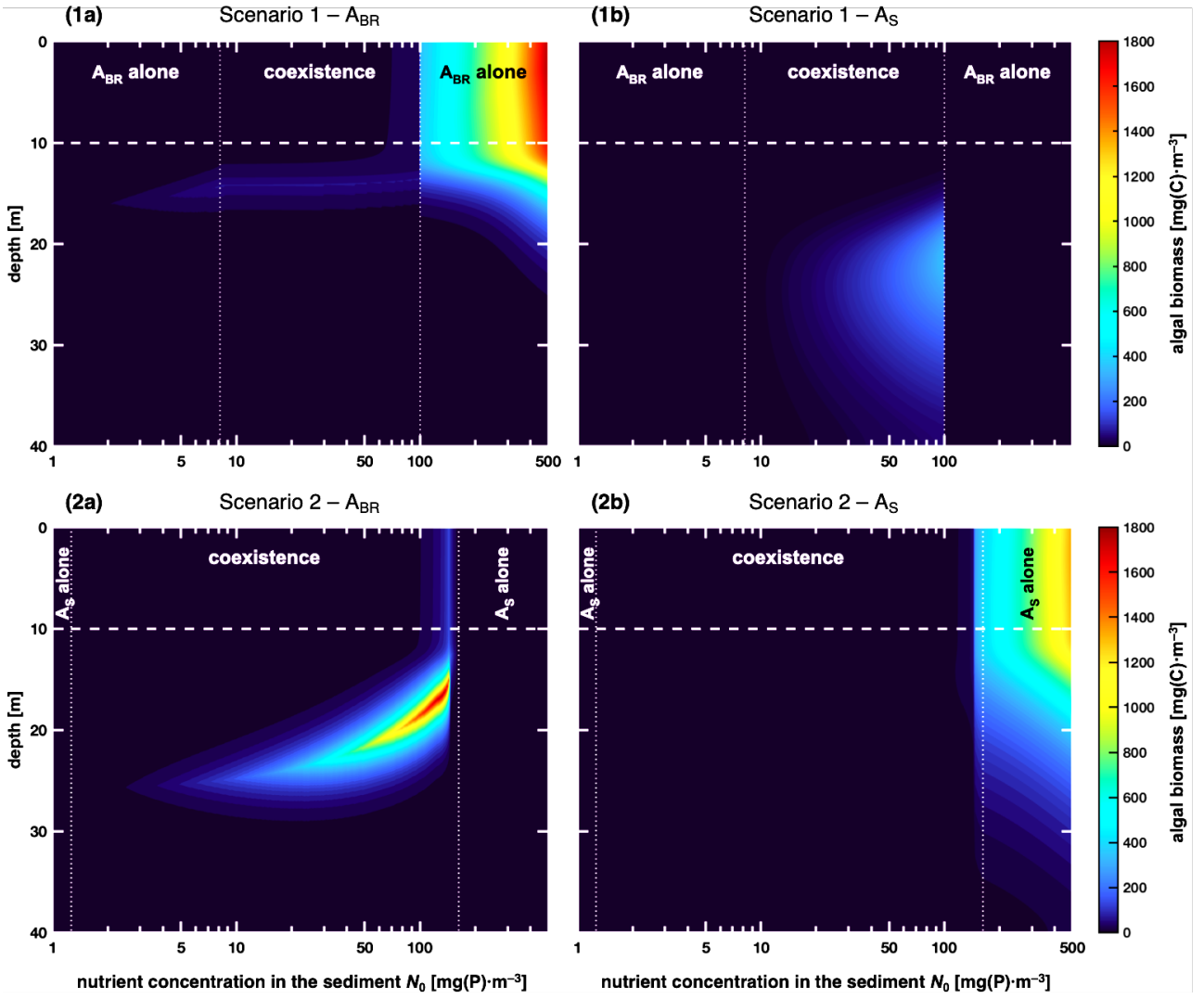


Figure S3. Influence of the environmental gradient of nutrient concentration in the sediment (N_0) on biomass profiles of (a) A_{BR} and (b) A_S for (1) Scenario 1 (upper row) and (2) Scenario 2 (lower row). In each contour plot, the vertical axis is the depth [m] from surface to the bottom of the water column, the horizontal axis is the gradient of N_0 [$\text{mg(P)} \cdot \text{m}^{-3}$], and the color scale indicates the local algal biomass concentration [$\text{mg(C)} \cdot \text{m}^{-3}$]. The horizontal dashed white line delineates the thermocline (separation between epilimnion and hypolimnion, fixed at 10 m), while vertical dotted white lines indicate the boundary between different system states. Parameters are taken from Tab. 1, with $D_h = 1$ [$\text{m}^2 \cdot \text{day}^{-1}$] and $a_{bg} = 0.2$ [m^{-1}].

S2. Competition outcomes in the two-parameter space of algal nutrient quotas

S2.1. Scenario 1

In the two-parameter plane defined by the P:C ratio of both species (q_{BR} , q_S) (Fig. S4(a)), a critical threshold at $q_{BR} \approx 0.0056$ [mg(P)·[mg(C)]⁻¹], denoted as q_{BR}^* , marks the boundary between competitive exclusion domains for lower and coexistence states for higher values of q_{BR} . For $q_{BR} > q_{BR}^*$, the coexistence state is characterized by A_S forming a DCM and A_{BR} forming an ECM for low q_S or a DCM for intermediate to high q_S values. For $q_{BR} < q_{BR}^*$, A_S is competitively excluded and A_{BR} forms an ECM. A bistability domain emerges for values of q_{BR} between q_{BR}^* and approximately 0.002 [mg(P)·[mg(C)]⁻¹], provided that $q_S \gtrsim 0.07$ [mg(P)·[mg(C)]⁻¹]. Within this region, the system can settle into one of two alternative steady states: either a single species state with A_{BR} forming an ECM, or a coexistence equilibrium with A_{BR} forming an ECM and A_S a DCM or both species forming a DCM for values of q_{BR} close to q_{BR}^* . Overall, the results suggest that minor variations in initial conditions or nutrient availability can lead to drastically different outcomes, highlighting the complex role of P:C ratio in phytoplankton competition and community composition.

S2.2. Scenario 2

In the two-parameter plane (q_{BR} , q_S) (Fig. S4(b)), the competition outcomes differ significantly from those observed for Scenario 1 (Fig. S4(a)). Now, a critical threshold at $q_S \approx 0.026$ [mg(P)·[mg(C)]⁻¹], denoted as q_S^* , delineates competitive exclusion domains for lower and coexistence states for higher values of q_S . In other words, q_S^* marks the boundary between A_{BR} 's survival and non-survival. For $q_S < q_S^*$ and $q_{BR} \lesssim 0.008$ [mg(P)·mg(C)⁻¹], A_S outcompetes A_{BR} in the epilimnion. Notably, for $q_{BR} > 0.008$ [mg(P)·mg(C)⁻¹], alternative steady states emerge between the ECM of A_S and a coexistence state in the hypolimnion: if A_{BR} is initially the resident species, it stays in the epilimnion, but if A_S is the initial resident, both species coexist in deeper layers forming DCMs. For P:C ratios of A_S (q_S) above q_S^* , only coexistence is possible: the two species either coexist in the same or in distinct layers, depending on q_{BR} and q_S values. However, coexistence within deeper layers (both species forming DCMs) remains the primary domain in the two-parameter plane.

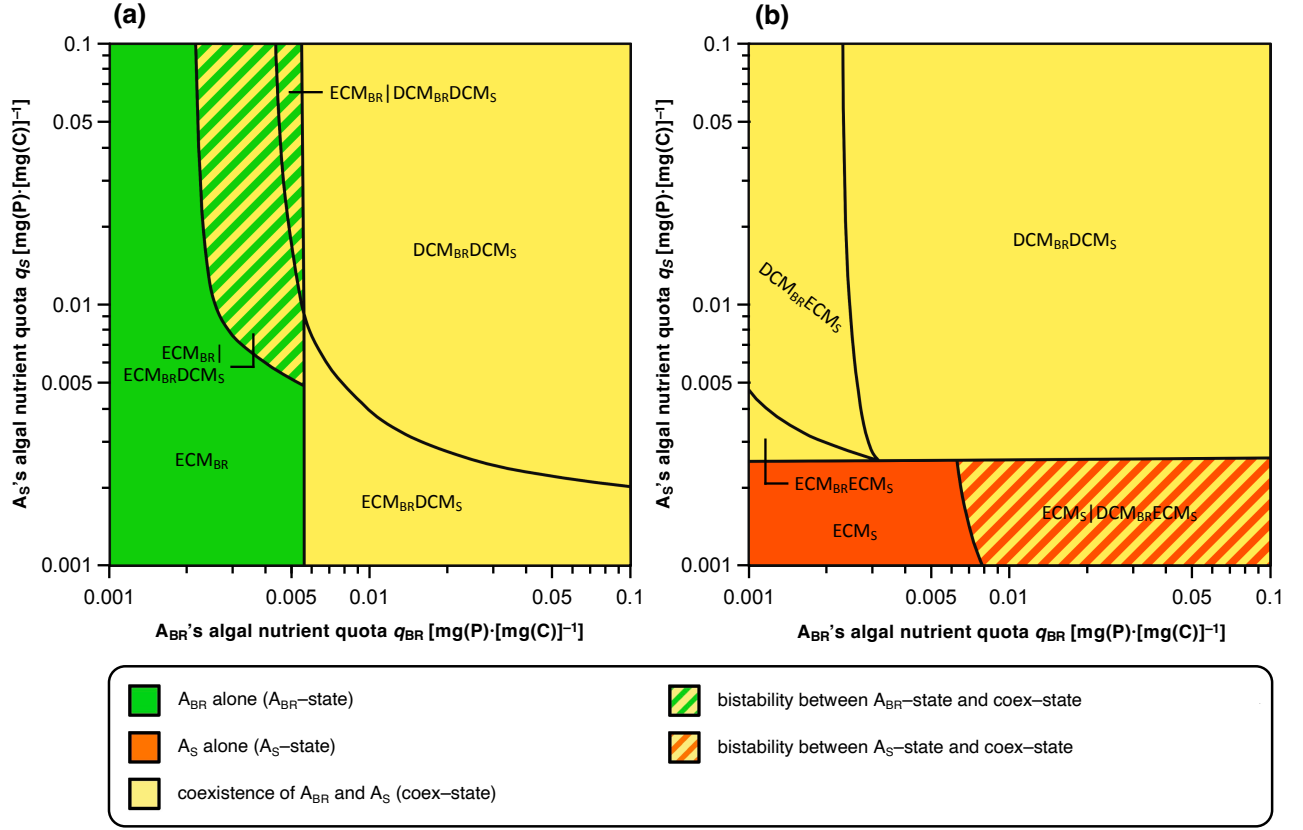


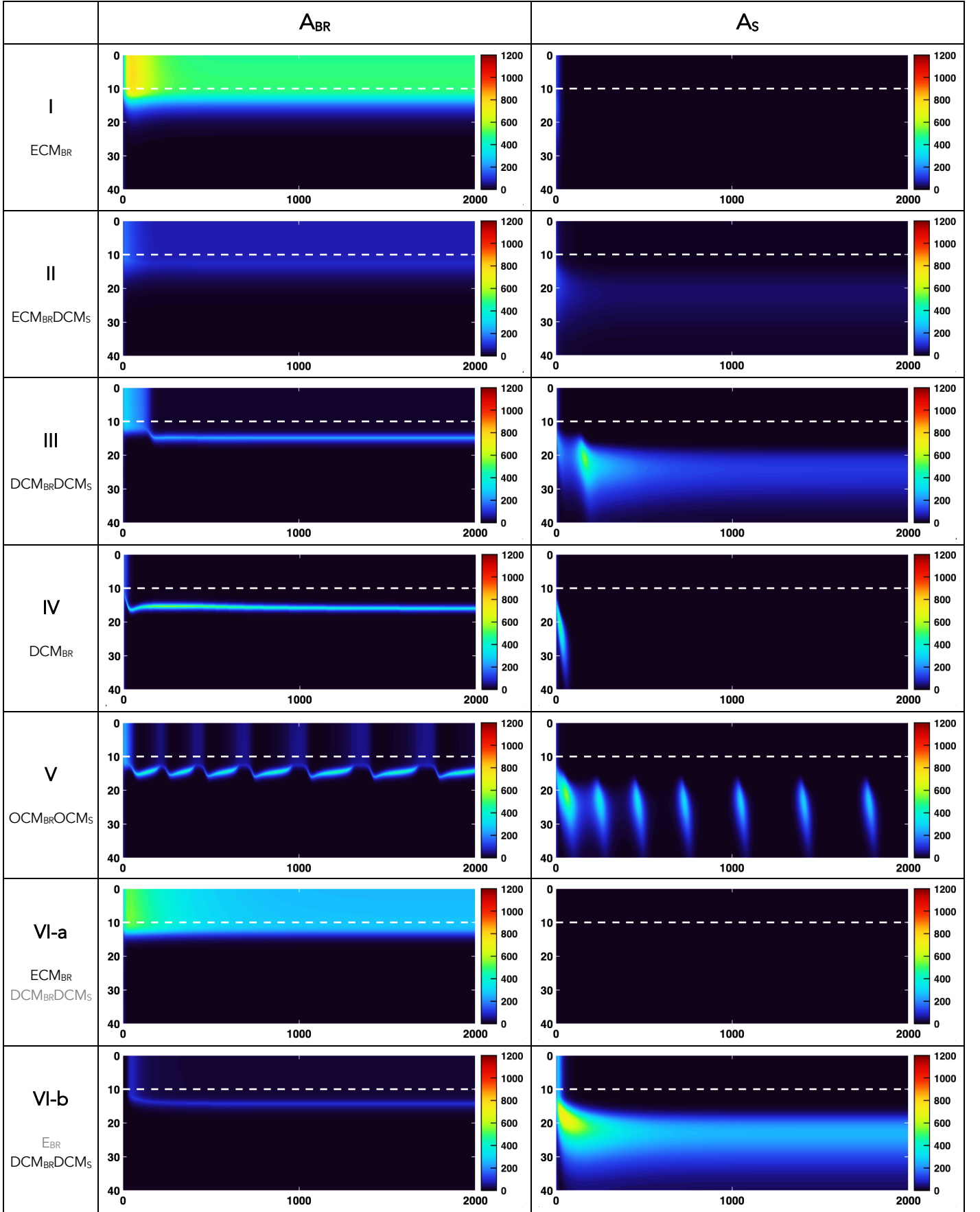
Figure S4. Bifurcation plots of competition outcomes in the two-parameter plane determined by the species-specific algal nutrient quotas (i.e., P:C ratios $[mg(P) \cdot [mg(C)]^{-1}]$) (q_{BR} , q_S) of A_{BR} and A_S , respectively. In subplot (a), A_{BR} is an I -sp. and A_S is an N -sp., with half-saturation constants set to $K_{BR} = 0.5 [mg(P) \cdot m^{-3}]$, $H_{BR} = 120 [\mu mol(photons) \cdot m^{-2} \cdot s^{-1}]$, $K_S = 3 [mg(P) \cdot m^{-3}]$, and $H_S = 20 [\mu mol(photons) \cdot m^{-2} \cdot s^{-1}]$. In subplot (b), A_{BR} is an N -sp. and A_S is an I -sp., with half-saturation constants set to $K_{BR} = 3 [mg(P) \cdot m^{-3}]$, $H_{BR} = 20 [\mu mol(photons) \cdot m^{-2} \cdot s^{-1}]$, $K_S = 0.5 [mg(P) \cdot m^{-3}]$, and $H_S = 120 [\mu mol(photons) \cdot m^{-2} \cdot s^{-1}]$. Other parameters are taken from Tab. 1. Colors stand for different competition outcomes (system states): green for A_{BR} -state, red for A_S -state, yellow for coex-state. Furthermore, the location of the biomass peak of each species is indicated by ECM = epilimnetic chlorophyll maximum; DCM = deep chlorophyll maximum. If two alternative system states are indicated, separated by a vertical bar, this indicates an area of bistability within which potentially both states can be reached, dependent on initial conditions.

S3. Examples for phytoplankton biomass profiles over depth and time

The simulations of vertical phytoplankton biomass distribution over time shown in Fig. S5 illustrate the transient dynamics for different system states observed in Fig. 4A(a) of the main article. A list of the examples shown in the different panels of Fig. S5 and a brief description of the observed system state is provided in Tab. S1.

Table S1. List and brief description of examples for biomass profiles over depth and time shown in Fig. S5.

figure	system state	description of system state
Fig. S5-I	A_{BR} -state	The BR species survives in the epilimnion (ECM) whereas the sinking species goes extinct.
Fig. S5-II	coex-state	Both species coexist in different part of the water column: the BR species in the epilimnion (ECM) and the sinking species in the hypolimnion (DCM).
Fig. S5-III	coex-state	The BR species initially blooms in the epilimnion and then establishes in the hypolimnion (DCM), while the sinking species after an initial biomass decline forms a DCM at even lower depth compared to A_{BR} .
Fig. S5-IV	A_{BR} -state	The BR species survives in the hypolimnion (DCM) whereas the sinking species goes extinct.
Fig. S5-V	oscillatory coex-state	The BR species biomass oscillates between the hypolimnion and the epilimnion, while the sinking species remains at the same depth in the hypolimnion, oscillating between low and high biomass levels.
Fig. S5-VIa and S5-VIb	[VIa] A_{BR} -state [VIb] A_S -state	[VIa] When the BR species is the resident, it survives in the epilimnion (DCM), preventing successful invasion of the sinking species. [VIb] Conversely, when the sinking species is the resident, both species coexist in the hypolimnion, the DCM of A_S being at a lower depth compared to A_{BR} .
Fig. S5-VIIa and S5-VIIb	[VIIa] A_{BR} -state [VIIb] oscillatory coex-state	[VIIa] When the BR species is the resident, it grows in the epilimnion (ECM), preventing successful invasion of the sinking species. [VIIb] Conversely, when the sinking species is the resident, both species coexist, ABR oscillating between ECM and DCM, A_S staying in the hypolimnion, showing oscillations between low and high biomass levels (similar to Fig. S5-V).
Fig. S5-VIIIa and S5-VIIIb	[VIIIa] A_{BR} -state (ECM) [VIIIb] A_{BR} -state (DCM)	Independent of who is the resident versus invader, the sinking species always goes extinct. [VIIIa] When the BR species is the resident, it forms an ECM, [VIIIb] when it is the invader, it settles in the hypolimnion (DCM).
Fig. S5-IX	coex-state	After a long transient phase, in which the BR species concentrates in the ECM and the sinking species is almost extinct, the sinking species shows oscillatory biomass shifts in the hypolimnion (DCM), whereas the BR species shows decreased biomass in the epilimnion and forms a thin DCM very close to the thermocline.



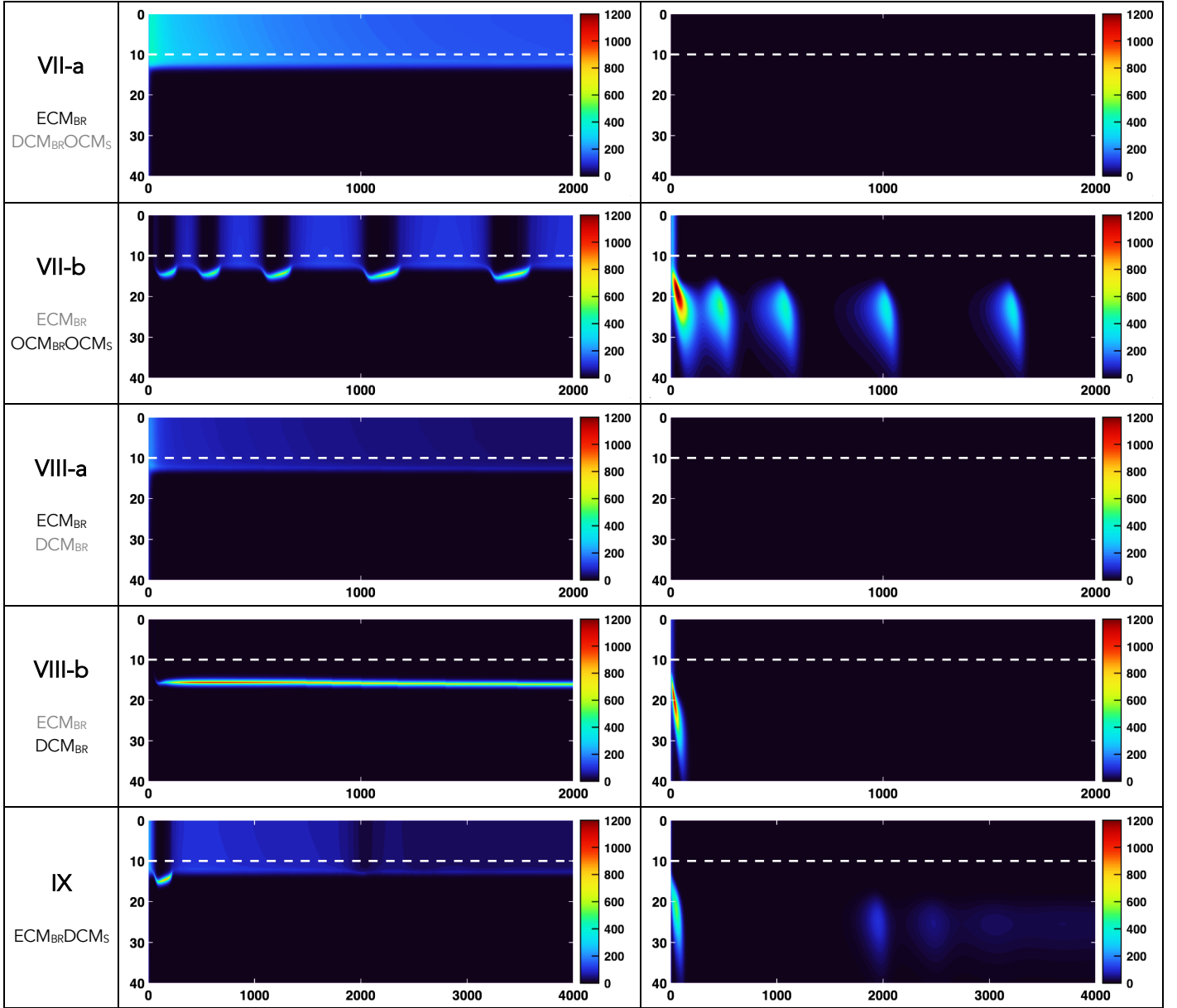


Figure S5. Examples of phytoplankton biomass profiles over depth (vertical axis, [m]) and time (horizontal axis, [day]) for different values of D_h and N_0 , illustrating different system states. The color scale indicates the algal biomass [$\text{mg(C)} \cdot \text{m}^{-3}$]. These profiles are obtained for Scenario 1 ($A_{BR} = I\text{-sp.}$, $A_S = N\text{-sp.}$), corresponding to the results shown in Fig. 4A(a). Convergence to a steady state is reached in all panels except panels (V) and (VII-b), where permanent oscillations are observed. Subpanels, labeled **a** and **b**, differ in the initial conditions: (a) A_{BR} is the resident and A_S the invader, (b) A_{BR} is the invader and A_S the resident. The horizontal dashed white line delineates the thermocline. Parameters are taken from Tab. 1, with: $D_h = 2.1$ (I), 2.9 (II), 0.5 (III), 0.3 (IV), 0.35 (V), 0.75 (VI), 0.27 (VII), 0.2 (VIII), 0.24 (IX) [$\text{m}^2 \cdot \text{day}^{-1}$]; $N_0 = 67$ (I), 14 (II), 80 (III), 30 (IV), 70 (V), 93 (VI), 125 (VII), 50 (VIII), 77 (IX) [$\text{mg(P)} \cdot \text{m}^{-3}$].

S4. Light and nutrient limitation along the water column

To assess resource limitation along the water column, we focused on Liebig's 'law of the minimum' (see Eq. 2 in the main article), implemented in species i 's growth rate as

$$g_i(N, I) = \mu_i \min \left\{ \frac{N}{K_i + N}, \frac{I}{H_i + I} \right\}$$

for $i \in \{\text{BR}, \text{S}\}$. We define the quantity

$$\lambda_i(z) = \frac{N}{K_i + N} - \frac{I}{H_i + I}$$

called the "level of resource limitation". If $\lambda_i(z) > 0$, then $g_i(N, I) = \mu_i \frac{I}{H_i + I}$ and species i 's growth at depth z is limited by light. If $\lambda_i(z) < 0$, then $g_i(N, I) = \mu_i \frac{N}{K_i + N}$ and species i 's growth at depth z is limited by nutrients. If $\lambda_i(z) = 0$, then species i 's growth at depth z is co-limited by both resources. The following figures (Fig. S6 to S8) show the level of resource limitation for A_{BR} and A_{S} , over depth and along similar environmental gradients, as investigated in Fig. S1 to S3 (gradients of D_{h} , a_{bg} , and N_0), respectively. Additionally, Fig. S9 shows A_{BR} 's and A_{S} 's level of resource limitation over depth and time for the oscillatory chlorophyll maximum reported in Fig. 4.

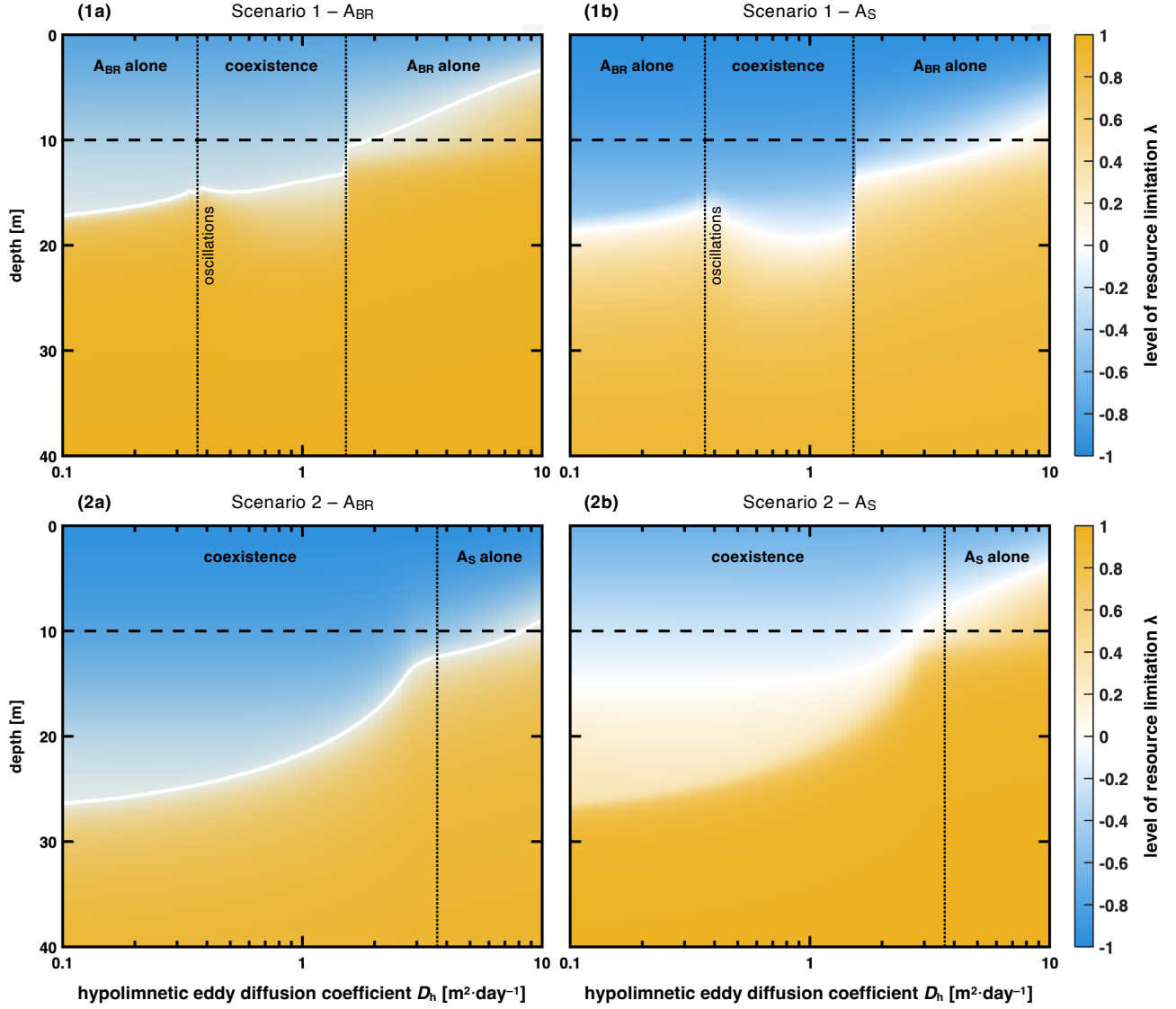


Figure S6. Influence of the environmental gradient of hypolimnetic eddy diffusion coefficient (D_h) on resource limitation of (a) A_{BR} and (b) A_S for (1) Scenario 1 (upper row) and (2) Scenario 2 (lower row). In each contour plot, the vertical axis is the depth [m] from surface to the bottom of the water column, the horizontal axis is the gradient of D_h [$\text{m}^2 \cdot \text{day}^{-1}$], and the color scale indicates the level of resource limitation (quantified by $\lambda_i(z)$, $i \in \{BR, S\}$). White areas indicate co-limitation ($\lambda_i = 0$), blue areas indicate nutrient limitation of algal growth ($\lambda_i < 0$), and yellow areas indicate light limitation of algal growth ($\lambda_i > 0$). The horizontal dashed grey line indicates the thermocline (separation between epilimnion and hypolimnion), while vertical dotted grey lines indicate boundaries between different system states. Parameters are taken from Tab. 1, with $a_0 = 0.2$ [m^{-1}] and $N_0 = 50$ [$\text{mg(P)} \cdot \text{m}^{-3}$].

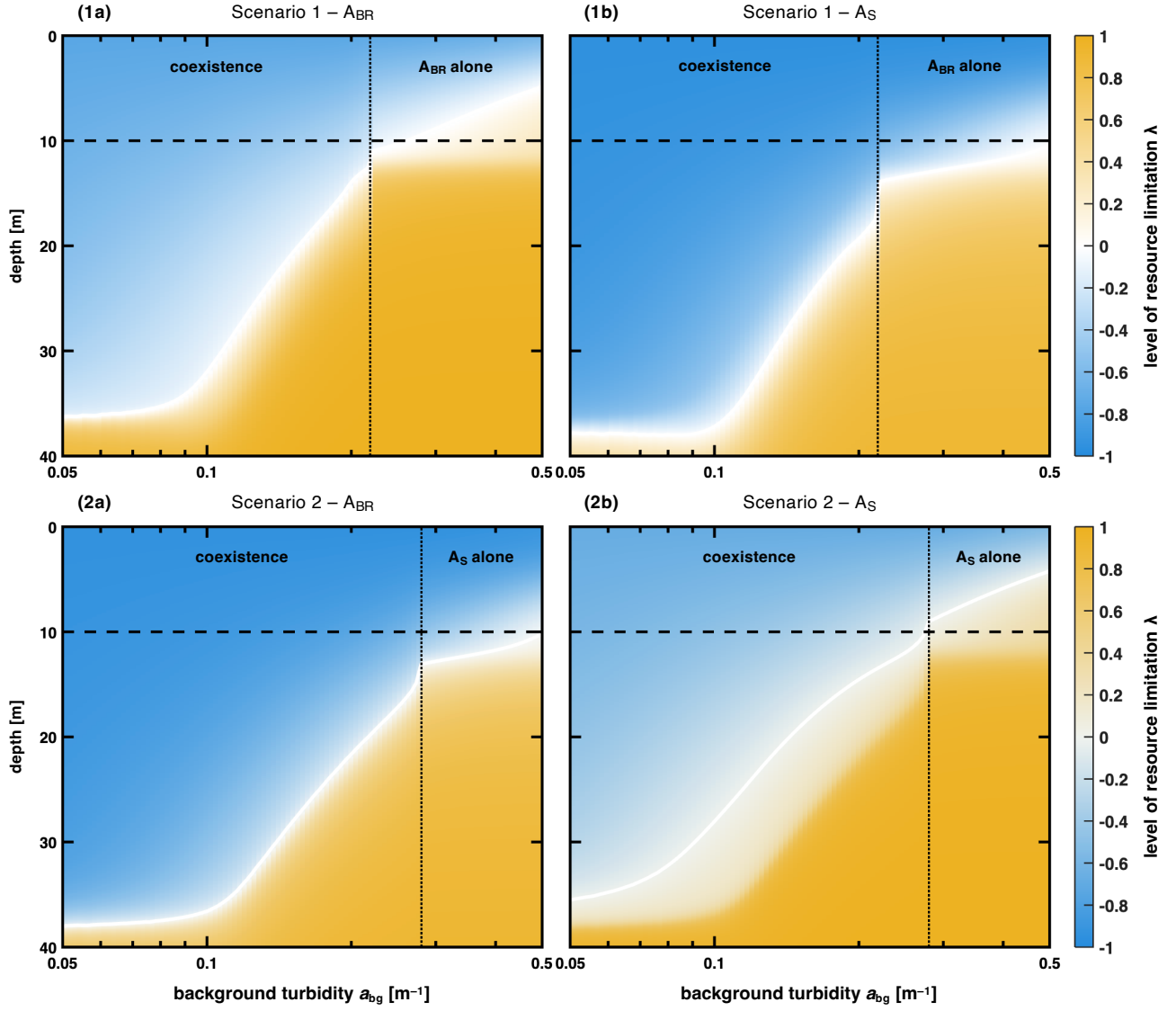


Figure S7. Influence of the environmental gradient of background turbidity (a_{bg}) on resource limitation of (a) A_{BR} and (b) A_S for (1) Scenario 1 (upper row) and (2) Scenario 2 (lower row). In each contour plot, the vertical axis is the depth [m], the horizontal axis is the gradient of a_{bg} [m^{-1}] from surface to the bottom of the water column, and the color scale indicates the level of resource limitation (quantified by $\lambda_i(z)$, $i \in \{BR, S\}$). White areas indicate co-limitation ($\lambda_i = 0$), blue areas indicate nutrient limitation of algal growth ($\lambda_i < 0$), and yellow areas indicate light limitation of algal growth ($\lambda_i > 0$). The horizontal dashed grey line delineates the thermocline (separation between epilimnion and hypolimnion), while vertical dotted grey lines indicate boundaries between different system states. Parameters are taken from Tab. 1, with $D_h = 1$ [$m^2 \cdot day^{-1}$] and $N_0 = 50$ [$mg(P) \cdot m^{-3}$].

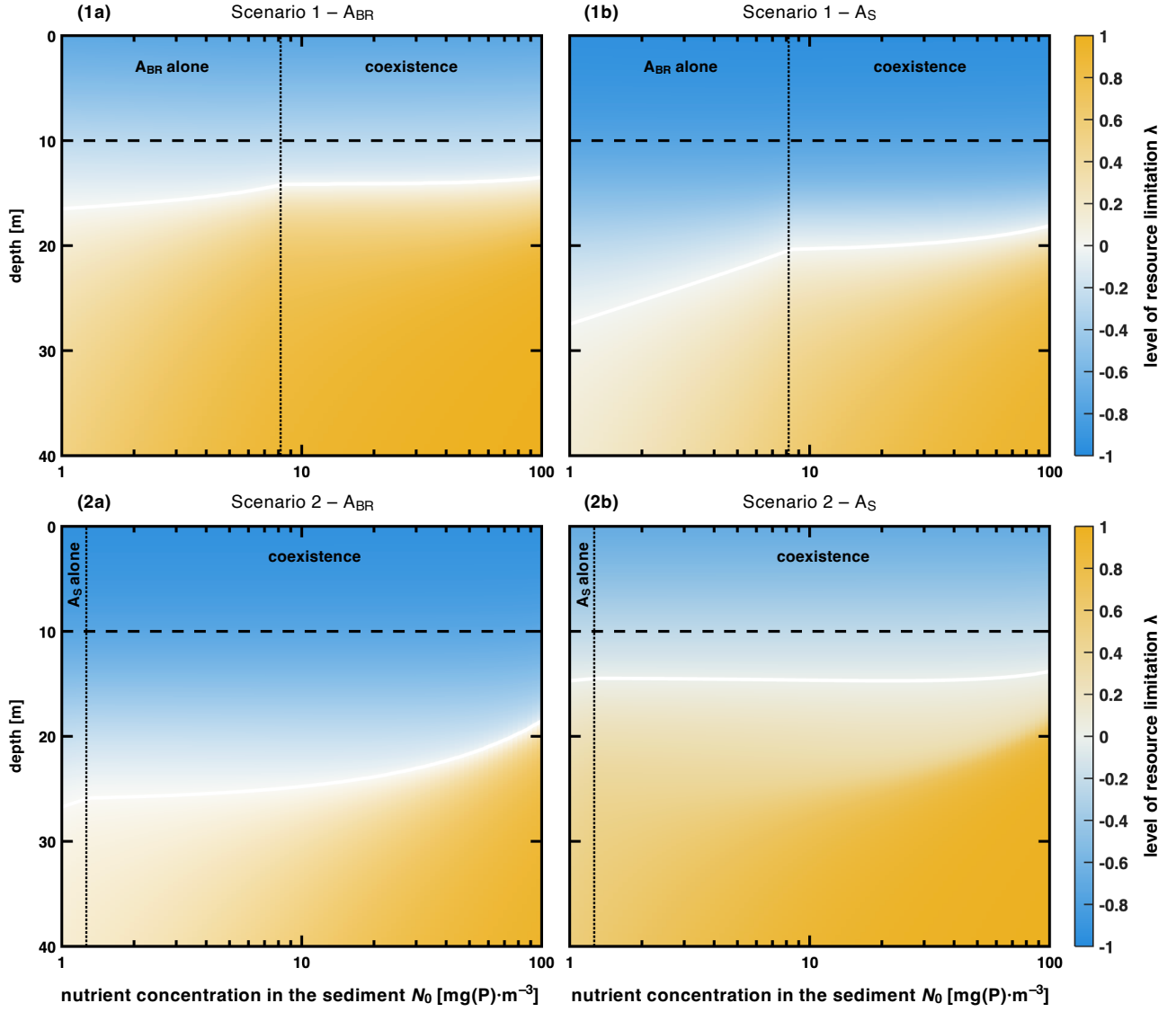


Figure S8. Influence of the environmental gradient of nutrient concentration in the sediment (N_0) on resource limitation of (a) A_{BR} and (b) A_S for (1) Scenario 1 (upper row) and (2) Scenario 2 (lower row). In each contour plot, the vertical axis is the depth [m] from surface to the bottom of the water column, the horizontal axis is the gradient of N_0 [$\text{mg(P)} \cdot \text{m}^{-3}$], and the color scale indicates the level of resource limitation (quantified by $\lambda_i(z)$, $i \in \{\text{BR}, \text{S}\}$). White areas indicate co-limitation ($\lambda_i = 0$), blue areas indicate nutrient limitation of algal growth ($\lambda_i < 0$), and yellow areas indicate light limitation of algal growth ($\lambda_i > 0$). The horizontal dashed grey line delineates the thermocline (separation between epilimnion and hypolimnion), while vertical dotted grey lines indicate boundaries between different system states. Parameters are taken from Tab. 1, with $D_h = 1$ [$\text{m}^2 \cdot \text{day}^{-1}$] and $a_{bg} = 0.2$ [m^{-1}].

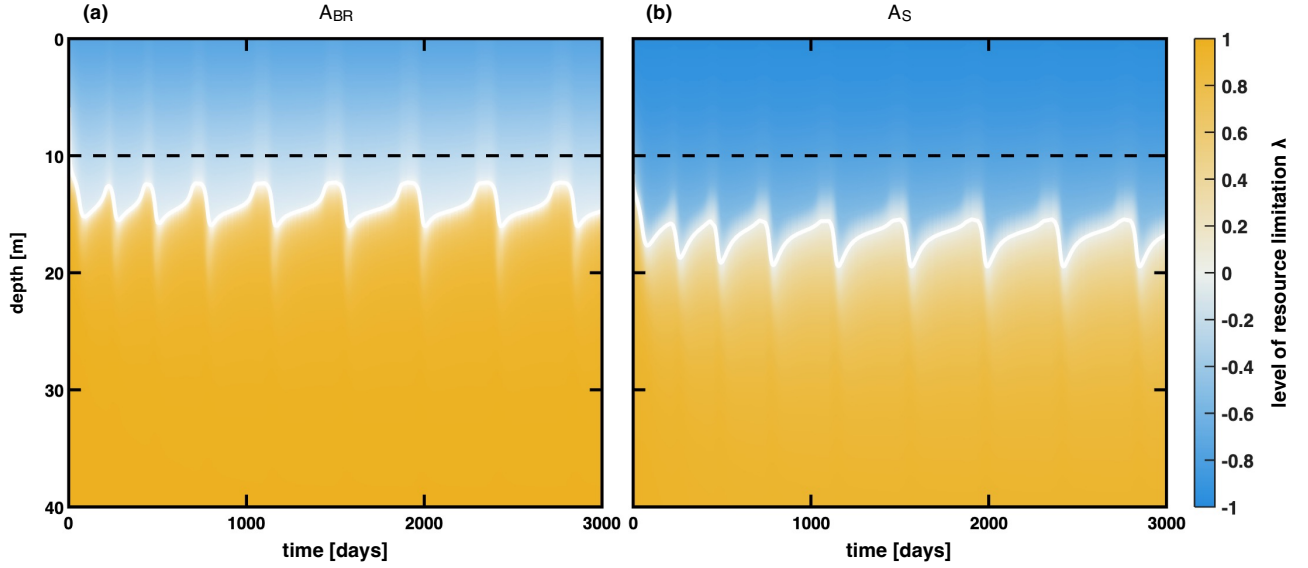


Figure S9. Resource limitation of **(a)** A_{BR} and **(b)** A_S for the oscillatory coexistence state reported in Fig. 5 of the main article. In each contour plot, the vertical axis is the depth [m] from surface to the bottom of the water column, the horizontal axis is the time [days], and the color scale indicates the level of resource limitation (quantified by $\lambda_i(z)$, $i \in \{BR, S\}$). White areas indicate co-limitation ($\lambda_i = 0$), blue areas indicate nutrient limitation of algal growth ($\lambda_i < 0$), and yellow areas indicate light limitation of algal growth ($\lambda_i > 0$). The horizontal dashed grey line delineates the thermocline (separation between epilimnion and hypolimnion). Parameters are taken from Tab. 1, with $D_h = 0.35 \text{ [m}^2 \cdot \text{day}^{-1}]$ and $N_0 = 70 \text{ [mg(P)} \cdot \text{m}^{-3}]$.

S5. Mathematical evidence of the competitive advantage provided by buoyancy regulation

The trait of buoyancy regulation can confer a competitive advantage, as algal cells can actively influence their vertical position along the water column and settle at the depth of optimal resource availability (namely, light and nutrient). Herein, we provide mathematical evidence for the competitive advantage of BR species over sinking taxa. Throughout this section, we assume, for the sake of simplicity, that $D(z)$ is constant along the water column (no stratification): $D(z) = D = \text{constant}$.

S5.1. Buoyancy-regulating velocity profile

Along the water column, light intensity decreases with depth while nutrient concentration increases with depth (as nutrients are assumed to diffuse from the sediment into the water column). In the model used to investigate competition between BR and S species (see Eq. 1 in the main article), we assume that the growth rate is based on Liebig's 'law of the minimum' and that resource uptake follows Monod kinetics, i.e.,

$$g_{\text{BR}}(z) = \mu_{\text{BR}} \min \left\{ \frac{N}{K_{\text{BR}} + N}, \frac{I}{H_{\text{BR}} + I} \right\}$$

Write henceforth $f_N = \frac{N}{K_{\text{BR}} + N}$ and $f_I = \frac{I}{H_{\text{BR}} + I}$ for the two Monod components. Using the relation $\min\{x, y\} = \frac{x+y-|x-y|}{2} = \frac{x+y-\sqrt{(x-y)^2}}{2}$, the growth rate is differentiable in z at every depth at which $f_N \neq f_I$, and one thus obtains

$$\begin{aligned} \frac{\partial g_{\text{BR}}}{\partial z} = \frac{\mu_{\text{BR}}}{2} & \left(\frac{H_{\text{BR}}}{(H_{\text{BR}} + I)^2} \frac{\partial I}{\partial z} + \frac{K_{\text{BR}}}{(K_{\text{BR}} + N)^2} \frac{\partial N}{\partial z} \right. \\ & \left. + \frac{\frac{H_{\text{BR}}}{(H_{\text{BR}} + I)^2} \frac{\partial I}{\partial z} - \frac{K_{\text{BR}}}{(K_{\text{BR}} + N)^2} \frac{\partial N}{\partial z}}{f_N - f_I} |f_N - f_I| \right) \end{aligned}$$

The two essential resources usually follow opposite gradients $\frac{\partial I}{\partial z} < 0$ and $\frac{\partial N}{\partial z} > 0$, thus f_N increases with depth, while f_I decreases with depth. Consequently, $g_{\text{BR}} = \mu_{\text{BR}} \min\{f_N, f_I\}$ is unimodal in z : it increases on the nutrient-limited interval ($f_N < f_I$) and decreases on the light-limited interval ($f_N > f_I$), reaching its maximum at the unique co-limitation depth $z^* \in [0, \ell]$ such that $f_N(z^*) = f_I(z^*)$. When the same resource limits growth throughout the entire water column, the maximum is reached at a boundary, i.e., $z^* = 0$ or $z^* = \ell$; we however assume $0 < z^* < \ell$ for the remainder of the section. Accordingly, A_{BR} moves vertically toward depth z^* : cells located above z^* sink toward it, while cells located below z^* rise toward it.

Because f_N (increasing) and f_I (decreasing) cross transversally at z^* , the growth rate has a concave corner there, and $\frac{\partial g_{BR}}{\partial z}$ jumps from the positive value $\mu_{BR} \frac{\partial f_N}{\partial z} \Big|_{z=z^*}$ to the negative value $\mu_{BR} \frac{\partial f_I}{\partial z} \Big|_{z=z^*}$. With the smooth formulation of $V_{BR}(z)$ (see Eq. 5 in the main article), we have $V_{BR}(z) > 0$ (downward) for $z < z^*$ and $V_{BR}(z) < 0$ (upward) for $z > z^*$; hence, the velocity reverses sign across z^* , and both branches drive biomass toward the depth of optimal growth, where biomass accumulates. For any finite $\kappa > 0$, the velocity is bounded: $|V_{BR}(z)| < v_{BR}$ for any z , with $|V_{BR}(z)| \rightarrow v_{BR}$ only in the limit $\kappa \rightarrow 0$. In particular, $0 < V_{BR}(z=0) < v_{BR}$ and $-v_{BR} < V_{BR}(z=\ell) < 0$, where v_{BR} is the maximal vertical velocity of A_{BR} .

Wherever g_{BR} is twice differentiable (i.e., for any $z \neq z^*$), differentiating the velocity yields

$$\frac{\partial V_{BR}}{\partial z} = \frac{v_{BR} \frac{\partial^2 g_{BR}}{\partial z^2} \kappa}{\left(\left| \frac{\partial g_{BR}}{\partial z} \right| + \kappa \right)^2} \quad (S5.1)$$

so that $\frac{\partial V_{BR}}{\partial z}$ has the same sign as $\frac{\partial^2 g_{BR}}{\partial z^2}$. Hence, $\frac{\partial V_{BR}}{\partial z} \leq 0$ exactly where g_{BR} is concave. In particular, g_{BR} has a concave corner at z^* , where V_{BR} reverses sign from positive to negative — this sign reversal acts as a converging biomass front. The velocity exhibits a net boundary-to-boundary decrease: $V_{BR}(z=\ell) - V_{BR}(z=0) < 0$. We use this property and the associated sign structure in what follows. Under nutrient limitation ($g_{BR} = \mu_{BR} f_N$), Eq. S5.1 rewrites as

$$\frac{\partial V_{BR}}{\partial z} \Big|_{(N\text{-lim.})} = \frac{v_{BR} \mu_{BR} K_{BR} \left(\frac{\partial^2 N}{\partial z^2} (K_{BR} + N) - 2 \left(\frac{\partial N}{\partial z} \right)^2 \right) \kappa}{(K_{BR} + N)^3 \left(\left| \frac{\mu_{BR} K_{BR}}{(K_{BR} + N)^2} \frac{\partial N}{\partial z} \right| + \kappa \right)^2}$$

and under light limitation ($g_{BR} = \mu_{BR} f_I$), it rewrites as

$$\frac{\partial V_{BR}}{\partial z} \Big|_{(I\text{-lim.})} = \frac{v_{BR} \mu_{BR} H_{BR} \left(\frac{\partial^2 I}{\partial z^2} (H_{BR} + I) - 2 \left(\frac{\partial I}{\partial z} \right)^2 \right) \kappa}{(H_{BR} + I)^3 \left(\left| \frac{\mu_{BR} H_{BR}}{(H_{BR} + I)^2} \frac{\partial I}{\partial z} \right| + \kappa \right)^2}$$

S5.2. Reduced loss through the sediment boundary

We now show that the buoyancy regulation strategy yields an initial biomass advantage over the sinking strategy. It emerges from the fact that the sinking species loses biomass through the bottom of the water column, whereas the BR species avoids this loss by shifting its vertical position. Assume

that A_{BR} and A_S have the same physiological parameters, so that $g_{BR}(z) = g_S(z)$ and $m_{BR} = m_S$. The only difference between the two species is the vertical movement strategy. Biomass A_{BR} satisfies

$$\frac{\partial A_{BR}}{\partial t} = (g_{BR}(z) - m_{BR})A_{BR} - \frac{\partial}{\partial z} [V_{BR}A_{BR}] + \frac{\partial}{\partial z} \left[D \frac{\partial A_{BR}}{\partial z} \right] \quad (S5.2)$$

while biomass A_S satisfies

$$\frac{\partial A_S}{\partial t} = (g_S(z) - m_S)A_S - v_S \frac{\partial A_S}{\partial z} + \frac{\partial}{\partial z} \left[D \frac{\partial A_S}{\partial z} \right] \quad (S5.3)$$

The total biomasses are defined as

$$B_{BR}(t) = \int_0^\ell A_{BR}(z, t) dz \quad \text{and} \quad B_S(t) = \int_0^\ell A_S(z, t) dz$$

Writing the total flux $J_i = V_i A_i - D \frac{\partial A_i}{\partial z}$ and integrating Eq. S5.2 and S5.3 over the water column gives

$$\frac{dB_{BR}}{dt} = \int_0^\ell (g_{BR}(z) - m_{BR})A_{BR}(z, t) dz - [J_{BR}]_0^\ell$$

and

$$\frac{dB_S}{dt} = \int_0^\ell (g_S(z) - m_S)A_S(z, t) dz - [J_S]_0^\ell$$

The boundary conditions impose that the surface is closed for species' biomass, i.e., $J_i(z=0) = 0$, whereas the bottom is a convective outflow-only boundary, i.e., biomass leaves the water column only where the local velocity is directed downward and none enter from the sediments, thus $J_i(z=\ell) = \max\{V_i(z=\ell), 0\}A_i(z=\ell)$. For the sinking species, $V_S(z) = v_S > 0$ everywhere along the water column, thus $J_S(z=\ell) = v_S A_S(z=\ell)$. Hence,

$$\frac{dB_{BR}}{dt} = \int_0^\ell (g_{BR}(z) - m_{BR})A_{BR}(z, t) dz - \max\{V_{BR}(z=\ell), 0\} A_{BR}(z=\ell, t)$$

and

$$\frac{dB_S}{dt} = \int_0^\ell (g_S(z) - m_S)A_S(z, t) dz - v_S A_S(z=\ell, t)$$

Now, consider an initially empty water column, to which a small amount of both species is added uniformly:

$$A_{BR}(z, t=0) = A_S(z, t=0) = \frac{\hat{B}}{\ell} > 0$$

At $t=0$, both species have the same total biomass: $B_{BR}(t=0) = B_S(t=0) = \hat{B}$. Therefore, the initial difference in total biomass growth is determined only by the difference in bottom losses; thus at $t=0$,

$$\left. \frac{dB_{BR}}{dt} \right|_{t=0} - \left. \frac{dB_S}{dt} \right|_{t=0} = v_S \frac{\hat{B}}{\ell} - \max\{V_{BR}(z=\ell), 0\} \frac{\hat{B}}{\ell}$$

Using a first-order Taylor expansion over the small time interval dt ,

$$B_i(t = dt) = B_i(t = 0) + \left. \frac{dB_i}{dt} \right|_{t=0} dt + \mathcal{O}(dt^2)$$

we obtain

$$B_{BR}(t = dt) - B_S(t = dt) = (v_S - \max\{V_{BR}(z = \ell), 0\}) \frac{\hat{B}}{\ell} dt + \mathcal{O}(dt^2)$$

If $v_{BR} = v_S$, then $\max\{V_{BR}(z = \ell), 0\} \leq |V_{BR}(z = \ell)| < v_S$, yielding

$$B_{BR}(t = dt) > B_S(t = dt)$$

Thus, the BR species has an initial biomass advantage over the sinking species. In particular, when the optimal depth z^* is located in the interior of the water column ($0 < z^* < \ell$), then the BR cells located near the bottom of the water column move upward. It immediately gives $\max\{V_{BR}(z = \ell), 0\} = 0$, leading to

$$B_{BR}(t = dt) - B_S(t = dt) = v_S \frac{\hat{B}}{\ell} dt + \mathcal{O}(dt)$$

This mechanism arises from the convective outflow-only boundary condition together with the upward velocity of the BR species near the sediment layer.

S5.3. Production-aligned redistribution

We now show that the advantage of the BR species over the sinking species is also driven by the redistribution of the BR species' biomass toward alignment with the production layer. Expanding the advection term in Eq. S5.2 yields

$$\frac{\partial A_{BR}}{\partial t} = (g_{BR}(z) - m_{BR})A_{BR} - V_{BR} \frac{\partial A_{BR}}{\partial z} - \frac{\partial V_{BR}}{\partial z} A_{BR} + \frac{\partial}{\partial z} \left[D \frac{\partial A_{BR}}{\partial z} \right]$$

The term $-\frac{\partial V_{BR}}{\partial z} A_{BR}$ corresponds to the local compression/dilution of biomass due to the spatially-varying velocity profile. Define the gross production rate of species i as

$$\Pi_i(t) = \int_0^\ell g_i(z) A_i(z, t) dz$$

At $t = 0$, it holds $\Pi_{BR}(t = 0) = \Pi_S(t = 0)$. In order to compare their subsequent evolution in a common resource field, consider the advective component of $\frac{d\Pi_i}{dt}$, which we denote $\left. \frac{d\Pi_i}{dt} \right|_{adv}$. After an integration by parts, we obtain

$$\left. \frac{d\Pi_i}{dt} \right|_{adv} = - \int_0^\ell g_i(z) \frac{\partial}{\partial z} [V_i A_i] dz = \int_0^\ell \frac{\partial g_i}{\partial z} V_i A_i dz - [g_i V_i A_i]_0^\ell$$

For the BR species, we specifically have

$$- [g_{BR} V_{BR} A_{BR}]_0^\ell \geq 0$$

and

$$\int_0^\ell \frac{\partial g_{\text{BR}}}{\partial z} V_{\text{BR}} A_{\text{BR}} dz = v_{\text{BR}} \int_0^\ell \underbrace{\frac{\left(\frac{\partial g_{\text{BR}}}{\partial z}\right)^2}{\left|\frac{\partial g_{\text{BR}}}{\partial z}\right| + \kappa}}_{\geq 0} A_{\text{BR}} dz \geq 0$$

thus $\left.\frac{d\Pi_{\text{BR}}}{dt}\right|_{\text{adv}} \geq 0$ and buoyancy regulation always transports biomass up the growth-rate gradient, toward the optimal depth z^* . For the sinking species, however, the sign of the term $v_{\text{S}} \int_0^\ell \frac{\partial g_{\text{S}}}{\partial z} A_{\text{S}} dz$ depends on the sign of $\frac{\partial g_{\text{S}}}{\partial z}$, which precisely shifts along the water column. Consequently, buoyancy regulation feeds gross production by locally concentrating biomass where g_{BR} is greatest, while ensuring no greater loss of bottom biomass.

S5.4. Equilibrium advantage

We now show that, at equilibrium and in a common resource field, the buoyancy regulation strategy confers a higher effective growth coefficient than the sinking strategy, provided that both species have the same physiological parameters (the only difference being the ability to adapt the vertical velocity of the BR species). Recall the reaction-advection-diffusion differential equation governing algal biomass of species i (with $i \in \{\text{BR}, \text{S}\}$):

$$\frac{\partial A_i}{\partial t} = (g_i(z) - m_i)A_i - \frac{\partial}{\partial z} [V_i A_i] + \frac{\partial}{\partial z} \left[D \frac{\partial A_i}{\partial z} \right] \quad (\text{S5.4})$$

Define $F_i(z) = \int_0^z \frac{V_i(x)}{2D} dx$ and let $A_i(z, t) = e^{F_i(z)} \psi_i(z, t)$. At equilibrium, Eq. S5.4 thus becomes

$$(g_i(z) - m_i)e^{F_i} \psi_i - \frac{\partial}{\partial z} [V_i e^{F_i} \psi_i] + \frac{\partial}{\partial z} \left[D \frac{\partial}{\partial z} [e^{F_i} \psi_i] \right] = 0$$

which, after calculations and simplifications, gives

$$(g_i(z) - m_i)\psi_i - \frac{V_i^2}{4D}\psi_i - \frac{1}{2} \frac{\partial V_i}{\partial z} \psi_i + D \frac{\partial^2 \psi_i}{\partial z^2} = 0 \quad (\text{S5.5})$$

Reorganizing Eq. S5.5 yields

$$\left(g_i(z) - m_i - \frac{V_i^2}{4D} - \frac{1}{2} \frac{\partial V_i}{\partial z} \right) \psi_i + D \frac{\partial^2 \psi_i}{\partial z^2} = 0 \quad (\text{S5.6})$$

For the BR species, Eq. S5.6 rewrites as

$$\underbrace{\left(g_{\text{BR}}(z) - m_{\text{BR}} - \frac{V_{\text{BR}}^2}{4D} - \frac{1}{2} \frac{\partial V_{\text{BR}}}{\partial z} \right)}_{= \eta_{\text{BR}}(z)} \psi_{\text{BR}} + D \frac{\partial^2 \psi_{\text{BR}}}{\partial z^2} = 0$$

and for the sinking species, it rewrites as

$$\underbrace{\left(g_S(z) - m_S - \frac{v_S^2}{4D}\right)}_{=\eta_S(z)} \psi_S + D \frac{\partial^2 \psi_S}{\partial z^2} = 0$$

Assuming both species have the same maximal growth rate, the same resource requirements, and the same mortality, we get $g_{BR}(z) = g_S(z)$, which results in

$$\eta_{BR}(z) - \eta_S(z) = \left(g_{BR}(z) - m_{BR} - \frac{V_{BR}^2}{4D} - \frac{1}{2} \frac{\partial V_{BR}}{\partial z}\right) - \left(g_S(z) - m_S - \frac{v_S^2}{4D}\right) = \underbrace{\frac{v_S^2 - V_{BR}^2}{4D}}_{(1)} - \underbrace{\frac{1}{2} \frac{\partial V_{BR}}{\partial z}}_{(2)}$$

at any depth $z \neq z^*$. The term (1) is non-negative everywhere since $|V_{BR}| \leq v_{BR} = v_S$. The term (2) is non-negative wherever $\frac{\partial V_{BR}}{\partial z} \leq 0$, i.e., wherever g_{BR} is concave. Hence, $\eta_{BR}(z) - \eta_S(z) \geq 0$ on every depth interval where g_{BR} is concave. Additionally,

$$\int_0^\ell (\eta_{BR}(z) - \eta_S(z)) dz = \int_0^\ell \underbrace{\frac{v_S^2 - V_{BR}^2}{4D}}_{>0} dz + \underbrace{\frac{1}{2} (V_{BR}(z=0) - V_{BR}(z=\ell))}_{>0} > 0$$

since $V_{BR}(z=0) > 0 > V_{BR}(z=\ell)$. Thus, at equilibrium and in a common resource field, the depth-averaged effective growth coefficient of the BR species exceeds that of the sinking species along the water column. However, recall that assessing (un)successful invasion would require computation of the principal eigenvalue, which is not done herein.

S5.5. Sinking as an effective additional biomass loss

It can also be shown that the washout due to sinking can be interpreted as a loss of biomass for the phytoplankton species in question (RYABOV, RUDOLF, and BLASIUS, 2010). Starting from

$$\frac{\partial A_S}{\partial t} = (g_S(z) - m_S) A_S - v_S \frac{\partial A_S}{\partial z} + \frac{\partial}{\partial z} \left[D \frac{\partial A_S}{\partial z} \right]$$

the substitution $A_S = e^{\frac{v_S z}{2D}} \phi_S$ provides

$$\frac{\partial \phi_S}{\partial t} = \left(g_S(z) - m_S - \frac{v_S^2}{4D}\right) \phi_S + D \frac{\partial^2 \phi_S}{\partial z^2}$$

Since $e^{\frac{v_S z}{2D}} > 0$, then at each fixed depth, A_S and ϕ_S increase and decrease simultaneously over time. Thus, the advection term acts as an additional, spatially-uniform loss term for the sinking species. Sinking lowers the effective net growth rate everywhere along the water column by $\frac{v_S^2}{4D}$. This interior effective loss is distinct from — and additional to — the irreversible loss of biomass at the bottom of the water column due to sedimentation (i.e., the convective bottom flux of Sect. S5.2).

S5.6. Synthesis

Overall, the competitive advantage provided by buoyancy regulation over pure sinking relies on two reinforcing mechanisms:

- Concentration toward the optimal growth depth due to buoyancy regulation (BR sp.): The converging velocity field continuously drives A_{BR} 's biomass toward the depth z^* where the growth rate $g_{BR}(z)$ is maximal.
- Effective additional biomass loss due to sinking (S sp.): Sinking imposes a uniform loss on A_S 's effective net growth at every depth, together with the irreversible loss of biomass through sedimentation at the bottom of the water column.

S6. ‘Method of Lines’ approach for solving PDEs

The ‘Method of Lines’ approach for solving PDEs, relying on a finite volume method, consists in two steps: (i) spatial differential and integral operators are discretized on a n -point spatial grid, then (ii) the resulting system of ODEs is numerically integrated in time (LUCAS *et al.*, 1998; SHARPLES and TETT, 1994). The numerical schemes for sinking phytoplankton species were fully described in HUISMAN and SOMMEIJER (2002), while those for buoyancy-regulating phytoplankton species were fully described in STOJSAVLJEVIC (2019). Herein, we combined and adapted the two sets of numerical schemes to deal with both sinking (S) and buoyancy-regulating (BR) species.

S6.1. General numerical technique

First, let $J_{A_{BR}}(z, t)$, $J_{A_S}(z, t)$, $J_N(z, t)$ be the A_{BR} ’s biomass flux, A_S ’s biomass flux, and nutrient flux, respectively. Thus, we have

$$\begin{cases} J_{A_{BR}}(z, t) = V_{BR}(z, t)A_{BR} - D(z)\frac{\partial A_{BR}}{\partial z} \\ J_{A_S}(z, t) = v_S A_S - D(z)\frac{\partial A_S}{\partial z} \\ J_N(z, t) = -D(z)\frac{\partial N}{\partial z} \end{cases}$$

verifying

$$\begin{cases} \frac{\partial A_{BR}}{\partial t} = (g_{BR}(N, I) - m_{BR})A_{BR} - \frac{\partial J_{A_{BR}}}{\partial z} \\ \frac{\partial A_S}{\partial t} = (g_S(N, I) - m_S)A_S - \frac{\partial J_{A_S}}{\partial z} \\ \frac{\partial N}{\partial t} = q_{BR}(rm_{BR} - g_{BR}(N, I))A_{BR} + q_S(rm_S - g_S(N, I))A_S - \frac{\partial J_N}{\partial z} \end{cases}$$

Spatial differential and integral operators are replaced by discrete approximations obtained on a n -point grid, which then results in a system of $3n$ ODEs: for $t > 0$,

$$\frac{d\mathbf{S}(t)}{dt} = \mathbf{F}(\mathbf{S}(t)) \quad (\text{S6.1})$$

with $\mathbf{S}(t) = \begin{pmatrix} \mathbf{A}_{BR}(t) \\ \mathbf{A}_S(t) \\ \mathbf{N}(t) \end{pmatrix} \in \mathbb{R}_+^{3n}$ where $\mathbf{A}_{BR}(t)$, $\mathbf{A}_S(t)$, $\mathbf{N}(t)$ are vectors of $\mathbb{R}_+^n$ containing the components

resulting from the discretization of A_{BR} , A_S , N , respectively. System (S6.1) can then be solved by numerical integration in time. Since system (Eq. S6.1) is stiff (HUISMAN and SOMMEIJER, 2002), we conducted the numerical integration in **MATLAB R2023b** software with solver **ode15s** for stiff

ODEs. We set `RelTol` to 10^{-8} and `AbsTol` to 10^{-10} , along with the `NonNegative` option for all variables.

Define the spatial grid of the one-dimensional water column ranging from $z = 0$ to $z = \ell$ as follows:

$$\begin{cases} u_0 = 0 \\ u_i = \left(i - \frac{1}{2}\right) \Delta z, \text{ for } i \in \llbracket 1, n \rrbracket \\ u_{n+1} = \ell \end{cases}$$

with $n = \frac{\ell}{\Delta z}$ (see Fig. S10). Consequently, we shall compute $A_{\text{BR}}(u_i, t)$, $A_{\text{S}}(u_i, t)$, $N(u_i, t)$ for all $i \in \llbracket 1, n \rrbracket$. For simplicity, we write $A_{\text{BR}}^{(i)} = A_{\text{BR}}(u_i, t)$, $A_{\text{S}}^{(i)} = A_{\text{S}}(u_i, t)$, $N^{(i)} = N(u_i, t)$, $J_{A_{\text{BR}}}^{(i)} = J_{A_{\text{BR}}}(u_i, t)$, $J_{A_{\text{S}}}^{(i)} = J_{A_{\text{S}}}(u_i, t)$, and $J_N^{(i)} = J_N(u_i, t)$, for any $i \in \{1, \dots, n\}$. We also denote $V_{\text{BR}}^{(i)} = V_{\text{BR}}\left(\frac{u_i + u_{i+1}}{2}, t\right)$ and $D^{(i)} = D\left(\frac{u_i + u_{i+1}}{2}, t\right)$, for any $i \in \{1, \dots, n\}$.

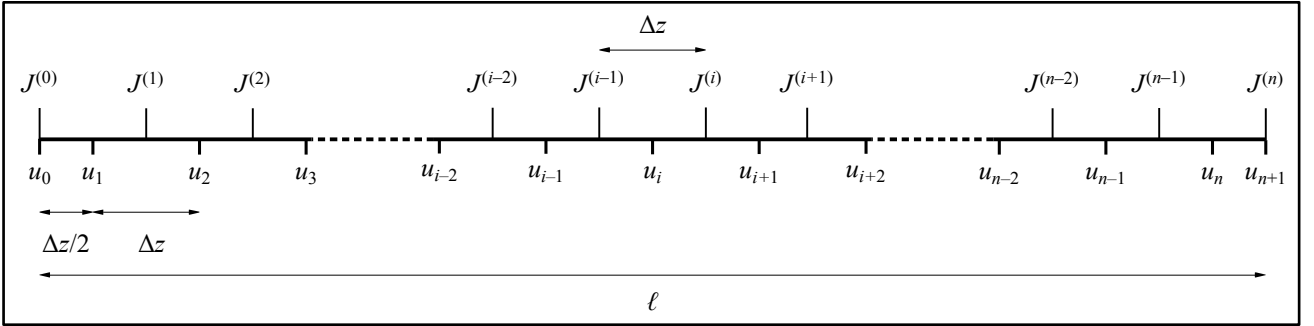


Figure S10. Spatial grid employed for the spatial discretization of the 1D water column.

For all $i \in \llbracket 1, n \rrbracket$, we approximate $\frac{\partial J_{A_{\text{BR}}}}{\partial z}$, $\frac{\partial J_{A_{\text{S}}}}{\partial z}$, and $\frac{\partial J_N}{\partial z}$ at the point $z = u_i$ as

$$\begin{cases} \left. \frac{\partial J_{A_{\text{BR}}}}{\partial z} \right|_{z=u_i, t} \approx \frac{J_{A_{\text{BR}}}^{(i)} - J_{A_{\text{BR}}}^{(i-1)}}{\Delta z} \\ \left. \frac{\partial J_{A_{\text{S}}}}{\partial z} \right|_{z=u_i, t} \approx \frac{J_{A_{\text{S}}}^{(i)} - J_{A_{\text{S}}}^{(i-1)}}{\Delta z} \\ \left. \frac{\partial J_N}{\partial z} \right|_{z=u_i, t} \approx \frac{J_N^{(i)} - J_N^{(i-1)}}{\Delta z} \end{cases}$$

We employ specific numerical schemes for computing spatial derivatives: a third-order upwind scheme for advection terms and a second-order symmetrical scheme for diffusion terms.

Recall that, at the surface of the water column, the boundary conditions impose

$$\begin{cases} J_{A_{BR}}^{(0)} = 0 \\ J_{A_S}^{(0)} = 0 \\ J_N^{(0)} = 0 \end{cases}$$

and, at the bottom of the water column, the boundary conditions impose

$$\begin{cases} \frac{\partial}{\partial z} [A_{BR}^{(n)}] = 0 \\ \frac{\partial}{\partial z} [A_S^{(n)}] = 0 \\ \frac{\partial}{\partial z} [N^{(n)}] = \xi(N_0 - N^{(n)}) \end{cases}$$

S6.2. Computation of the BR species' biomass flux derivative $\frac{\partial J_{A_{BR}}}{\partial z} \Big|_{z=u_i, t}$

For all $i \in \llbracket 1, n \rrbracket$, it holds

$$\begin{aligned} \frac{\partial J_{A_{BR}}}{\partial z} \Big|_{z=u_i, t} &= \frac{J_{A_{BR}}^{(i)} - J_{A_{BR}}^{(i-1)}}{\Delta z} \\ &= \frac{1}{\Delta z} \left((\{V_{BR} A_{BR}\}^{(i)} - \{V_{BR} A_{BR}\}^{(i-1)}) - \left(D^{(i)} \frac{\partial}{\partial z} [A_{BR}^{(i)}] - D^{(i-1)} \frac{\partial}{\partial z} [A_{BR}^{(i-1)}] \right) \right) \end{aligned}$$

According to the definition of $V_{BR}(z, t)$, which varies depending on the BR species' growth rate gradient, phytoplankton will then move up or down along the water column towards depths with better growth conditions. To apply the upwind scheme, we need to distinguish at each depth between cases when $V_{BR}^{(i)} > 0$ (downward movement) and cases when $V_{BR}^{(i)} < 0$ (upward movement). We thus denote

$$\begin{aligned} \mathbb{I}_{\{V_{BR}^{(i)} > 0\}} &= \begin{cases} 1 & \text{if } V_{BR}^{(i)} > 0 \\ 0 & \text{otherwise} \end{cases} \\ \mathbb{I}_{\{V_{BR}^{(i)} < 0\}} &= \begin{cases} 1 & \text{if } V_{BR}^{(i)} < 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

The implementation of the upwind scheme (FERZIGER and PERIC, 1999) for the advection term, encompassing the downward and upward movements, yields, for any $i \in \llbracket 2, n-1 \rrbracket$,

$$\begin{aligned} \{V_{BR} A_{BR}\}^{(i)} &= \frac{1}{6} \mathbb{I}_{\{V_{BR}^{(i)} > 0\}} V_{BR}^{(i)} \left(-A_{BR}^{(i-1)} + 5A_{BR}^{(i)} + 2A_{BR}^{(i+1)} \right) \\ &\quad + \frac{1}{6} \mathbb{I}_{\{V_{BR}^{(i)} < 0\}} V_{BR}^{(i)} \left(2A_{BR}^{(i)} + 5A_{BR}^{(i+1)} - A_{BR}^{(i+2)} \right) \\ &= [\text{downward movement}] + [\text{upward movement}] \end{aligned}$$

hence, for any $i \in \llbracket 3, n-2 \rrbracket$,

$$\begin{aligned}
& \{V_{\text{BR}}A_{\text{BR}}\}^{(i)} - \{V_{\text{BR}}A_{\text{BR}}\}^{(i-1)} \\
&= \frac{1}{6} \left(\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} > 0\}} V_{\text{BR}}^{(i-1)} A_{\text{BR}}^{(i-2)} \right. \\
&+ \left(-5\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} > 0\}} V_{\text{BR}}^{(i-1)} - 2\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} < 0\}} V_{\text{BR}}^{(i-1)} - \mathbb{I}_{\{V_{\text{BR}}^{(i)} > 0\}} V_{\text{BR}}^{(i)} \right) A_{\text{BR}}^{(i-1)} \\
&+ \left(-2\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} > 0\}} V_{\text{BR}}^{(i-1)} - 5\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} < 0\}} V_{\text{BR}}^{(i-1)} + 5\mathbb{I}_{\{V_{\text{BR}}^{(i)} > 0\}} V_{\text{BR}}^{(i)} \right. \\
&+ \left. 2\mathbb{I}_{\{V_{\text{BR}}^{(i)} < 0\}} V_{\text{BR}}^{(i)} \right) A_{\text{BR}}^{(i)} \\
&+ \left(\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} < 0\}} V_{\text{BR}}^{(i-1)} + 2\mathbb{I}_{\{V_{\text{BR}}^{(i)} > 0\}} V_{\text{BR}}^{(i)} + 5\mathbb{I}_{\{V_{\text{BR}}^{(i)} < 0\}} V_{\text{BR}}^{(i)} \right) A_{\text{BR}}^{(i+1)} \\
&\left. - \mathbb{I}_{\{V_{\text{BR}}^{(i)} < 0\}} V_{\text{BR}}^{(i)} A_{\text{BR}}^{(i+2)} \right)
\end{aligned}$$

The implementation of the symmetrical scheme for the diffusion term yields, for any $i \in \llbracket 3, n-2 \rrbracket$,

$$\frac{\partial}{\partial z} \left[A_{\text{BR}}^{(i)} \right] = \frac{A_{\text{BR}}^{(i+1)} - A_{\text{BR}}^{(i)}}{\Delta z}$$

hence, for any $i \in \llbracket 3, n-2 \rrbracket$,

$$\begin{aligned}
D^{(i)} \frac{\partial}{\partial z} \left[A_{\text{BR}}^{(i)} \right] - D^{(i-1)} \frac{\partial}{\partial z} \left[A_{\text{BR}}^{(i-1)} \right] &= \frac{1}{\Delta z} \left(D^{(i)} (A_{\text{BR}}^{(i+1)} - A_{\text{BR}}^{(i)}) - D^{(i-1)} (A_{\text{BR}}^{(i)} - A_{\text{BR}}^{(i-1)}) \right) \\
&= \frac{1}{\Delta z} \left(D^{(i-1)} A_{\text{BR}}^{(i-1)} - (D^{(i-1)} + D^{(i)}) A_{\text{BR}}^{(i)} + D^{(i)} A_{\text{BR}}^{(i+1)} \right)
\end{aligned}$$

Therefore, for any $i \in \llbracket 3, n-2 \rrbracket$,

$$\begin{aligned}
\frac{J_{A_{\text{BR}}}^{(i)} - J_{A_{\text{BR}}}^{(i-1)}}{\Delta z} &= \frac{1}{6\Delta z} \left(\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} > 0\}} V_{\text{BR}}^{(i-1)} A_{\text{BR}}^{(i-2)} \right. \\
&+ \left(-5\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} > 0\}} V_{\text{BR}}^{(i-1)} - 2\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} < 0\}} V_{\text{BR}}^{(i-1)} - \mathbb{I}_{\{V_{\text{BR}}^{(i)} > 0\}} V_{\text{BR}}^{(i)} \right) A_{\text{BR}}^{(i-1)} \\
&+ \left(-2\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} > 0\}} V_{\text{BR}}^{(i-1)} - 5\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} < 0\}} V_{\text{BR}}^{(i-1)} + 5\mathbb{I}_{\{V_{\text{BR}}^{(i)} > 0\}} V_{\text{BR}}^{(i)} \right. \\
&+ \left. 2\mathbb{I}_{\{V_{\text{BR}}^{(i)} < 0\}} V_{\text{BR}}^{(i)} \right) A_{\text{BR}}^{(i)} \\
&+ \left(\mathbb{I}_{\{V_{\text{BR}}^{(i-1)} < 0\}} V_{\text{BR}}^{(i-1)} + 2\mathbb{I}_{\{V_{\text{BR}}^{(i)} > 0\}} V_{\text{BR}}^{(i)} + 5\mathbb{I}_{\{V_{\text{BR}}^{(i)} < 0\}} V_{\text{BR}}^{(i)} \right) A_{\text{BR}}^{(i+1)} \\
&\left. - \mathbb{I}_{\{V_{\text{BR}}^{(i)} < 0\}} V_{\text{BR}}^{(i)} A_{\text{BR}}^{(i+2)} \right) - \frac{1}{\Delta z^2} \left(D^{(i-1)} A_{\text{BR}}^{(i-1)} - (D^{(i-1)} + D^{(i)}) A_{\text{BR}}^{(i)} + D^{(i)} A_{\text{BR}}^{(i+1)} \right)
\end{aligned}$$

At the surface of the water column, the boundary condition imposes $J_{A_{\text{BR}}}^{(0)} = 0$, and it holds

$$\{V_{\text{BR}}A_{\text{BR}}\}^{(1)} = \frac{1}{2} \mathbb{I}_{\{V_{\text{BR}}^{(1)} > 0\}} V_{\text{BR}}^{(1)} (A_{\text{BR}}^{(1)} + A_{\text{BR}}^{(2)}) + \frac{1}{6} \mathbb{I}_{\{V_{\text{BR}}^{(1)} < 0\}} V_{\text{BR}}^{(1)} (2A_{\text{BR}}^{(1)} + 5A_{\text{BR}}^{(2)} - A_{\text{BR}}^{(3)})$$

hence

$$\begin{aligned} \frac{J_{A_{BR}}^{(1)} - J_{A_{BR}}^{(0)}}{\Delta z} &= \frac{1}{2\Delta z} \mathbb{I}_{\{V_{BR}^{(1)} > 0\}} V_{BR}^{(1)} \left(A_{BR}^{(1)} + A_{BR}^{(2)} \right) + \frac{1}{6\Delta z} \mathbb{I}_{\{V_{BR}^{(1)} < 0\}} V_{BR}^{(1)} \left(2A_{BR}^{(1)} + 5A_{BR}^{(2)} - A_{BR}^{(3)} \right) \\ &\quad - \frac{1}{\Delta z^2} D^{(1)} \left(A_{BR}^{(2)} - A_{BR}^{(1)} \right) \end{aligned}$$

and

$$\begin{aligned} \frac{J_{A_{BR}}^{(2)} - J_{A_{BR}}^{(1)}}{\Delta z} &= \frac{1}{6\Delta z} \left(\left(-3\mathbb{I}_{\{V_{BR}^{(1)} > 0\}} V_{BR}^{(1)} - 2\mathbb{I}_{\{V_{BR}^{(1)} < 0\}} V_{BR}^{(1)} - \mathbb{I}_{\{V_{BR}^{(2)} > 0\}} V_{BR}^{(2)} \right) A_{BR}^{(1)} \right. \\ &\quad + \left(-5\mathbb{I}_{\{V_{BR}^{(1)} < 0\}} V_{BR}^{(1)} - 3\mathbb{I}_{\{V_{BR}^{(1)} > 0\}} V_{BR}^{(1)} + 2\mathbb{I}_{\{V_{BR}^{(2)} < 0\}} V_{BR}^{(2)} + 5\mathbb{I}_{\{V_{BR}^{(2)} > 0\}} V_{BR}^{(2)} \right) A_{BR}^{(2)} \\ &\quad + \left(\mathbb{I}_{\{V_{BR}^{(1)} < 0\}} V_{BR}^{(1)} + 5\mathbb{I}_{\{V_{BR}^{(2)} < 0\}} V_{BR}^{(2)} + 2\mathbb{I}_{\{V_{BR}^{(2)} > 0\}} V_{BR}^{(2)} \right) A_{BR}^{(3)} - \mathbb{I}_{\{V_{BR}^{(2)} < 0\}} V_{BR}^{(2)} A_{BR}^{(4)} \Big) \\ &\quad - \frac{1}{\Delta z^2} \left(D^{(1)} A_{BR}^{(1)} - (D^{(1)} + D^{(2)}) A_{BR}^{(2)} + D^{(2)} A_{BR}^{(3)} \right) \end{aligned}$$

At the bottom of the water column, the boundary condition imposes $\frac{\partial}{\partial z} \left[A_{BR}^{(n)} \right] = 0$ thus $A_{BR}^{(n+1)} = A_{BR}^{(n)}$

then

$$\{V_{BR} A_{BR}\}^{(n)} = \frac{1}{6} \mathbb{I}_{\{V_{BR}^{(n)} > 0\}} V_{BR}^{(n)} \left(-A_{BR}^{(n-1)} + 7A_{BR}^{(n)} \right) + \mathbb{I}_{\{V_{BR}^{(n)} < 0\}} V_{BR}^{(n)} A_{BR}^{(n)}$$

and it holds

$$\begin{aligned} \{V_{BR} A_{BR}\}^{(n-1)} &= \frac{1}{6} \mathbb{I}_{\{V_{BR}^{(n-1)} > 0\}} V_{BR}^{(n-1)} \left(-A_{BR}^{(n-2)} + 5A_{BR}^{(n-1)} + 2A_{BR}^{(n)} \right) \\ &\quad + \frac{1}{2} \mathbb{I}_{\{V_{BR}^{(n-1)} < 0\}} V_{BR}^{(n-1)} \left(A_{BR}^{(n-1)} + A_{BR}^{(n)} \right) \end{aligned}$$

hence, for $i = n$,

$$\begin{aligned} \frac{J_{A_{BR}}^{(n)} - J_{A_{BR}}^{(n-1)}}{\Delta z} &= \frac{1}{6\Delta z} \left(\mathbb{I}_{\{V_{BR}^{(n-1)} > 0\}} V_{BR}^{(n-1)} A_{BR}^{(n-2)} \right. \\ &\quad + \left(-5\mathbb{I}_{\{V_{BR}^{(n-1)} > 0\}} V_{BR}^{(n-1)} - 3\mathbb{I}_{\{V_{BR}^{(n-1)} < 0\}} V_{BR}^{(n-1)} - \mathbb{I}_{\{V_{BR}^{(n)} > 0\}} V_{BR}^{(n)} \right) A_{BR}^{(n-1)} \\ &\quad + \left(-2\mathbb{I}_{\{V_{BR}^{(n-1)} > 0\}} V_{BR}^{(n-1)} - 3\mathbb{I}_{\{V_{BR}^{(n-1)} < 0\}} V_{BR}^{(n-1)} + 7\mathbb{I}_{\{V_{BR}^{(n)} > 0\}} V_{BR}^{(n)} \right. \\ &\quad \left. \left. + 6\mathbb{I}_{\{V_{BR}^{(n)} < 0\}} V_{BR}^{(n)} \right) A_{BR}^{(n)} \right) - \frac{1}{\Delta z^2} D^{(n-1)} \left(A_{BR}^{(n-1)} - A_{BR}^{(n)} \right) \end{aligned}$$

and, for $i = n - 1$,

$$\begin{aligned}
& \frac{J_{A_{BR}}^{(n-1)} - J_{A_{BR}}^{(n-2)}}{\Delta z} \\
&= \frac{1}{6\Delta z} \left(\mathbb{I}_{\{V_{BR}^{(n-2)} > 0\}} V_{BR}^{(n-2)} A_{BR}^{(n-3)} \right. \\
&+ \left(-2\mathbb{I}_{\{V_{BR}^{(n-2)} < 0\}} V_{BR}^{(n-2)} - 5\mathbb{I}_{\{V_{BR}^{(n-2)} > 0\}} V_{BR}^{(n-2)} - \mathbb{I}_{\{V_{BR}^{(n-1)} > 0\}} V_{BR}^{(n-1)} \right) A_{BR}^{(n-2)} \\
&+ \left(-5\mathbb{I}_{\{V_{BR}^{(n-2)} < 0\}} V_{BR}^{(n-2)} - 2\mathbb{I}_{\{V_{BR}^{(n-2)} > 0\}} V_{BR}^{(n-2)} + 5\mathbb{I}_{\{V_{BR}^{(n-1)} > 0\}} V_{BR}^{(n-1)} \right. \\
&+ \left. 3\mathbb{I}_{\{V_{BR}^{(n-1)} < 0\}} V_{BR}^{(n-1)} \right) A_{BR}^{(n-1)} \\
&+ \left(\mathbb{I}_{\{V_{BR}^{(n-2)} < 0\}} V_{BR}^{(n-2)} + 2\mathbb{I}_{\{V_{BR}^{(n-1)} > 0\}} V_{BR}^{(n-1)} + 3\mathbb{I}_{\{V_{BR}^{(n-1)} < 0\}} V_{BR}^{(n-1)} \right) A_{BR}^{(n)} \\
&\left. - \frac{1}{\Delta z^2} \left(D^{(n-2)} A_{BR}^{(n-2)} - (D^{(n-2)} + D^{(n-1)}) A_{BR}^{(n-1)} + D^{(n-1)} A_{BR}^{(n)} \right) \right)
\end{aligned}$$

S6.3. Computation of the S species' biomass flux derivative $\left. \frac{\partial J_{A_S}}{\partial z} \right|_{z=u_i, t}$

For all $i \in \llbracket 1, n \rrbracket$, it holds

$$\begin{aligned}
\left. \frac{\partial J_{A_S}}{\partial z} \right|_{z=u_i, t} &= \frac{J_{A_S}^{(i)} - J_{A_S}^{(i-1)}}{\Delta z} \\
&= \frac{1}{\Delta z} \left((\{V_S A_S\}^{(i)} - \{V_S A_S\}^{(i-1)}) - \left(D^{(i)} \frac{\partial}{\partial z} [A_S^{(i)}] - D^{(i-1)} \frac{\partial}{\partial z} [A_S^{(i-1)}] \right) \right)
\end{aligned}$$

We only have a downward movement due to sinking, which results in the following upwind scheme, for any $i \in \llbracket 2, n-1 \rrbracket$,

$$\{V_S A_S\}^{(i)} = \frac{1}{6} v_S \left(-A_S^{(i-1)} + 5A_S^{(i)} + 2A_S^{(i+1)} \right)$$

hence, for any $i \in \llbracket 3, n-1 \rrbracket$,

$$\{V_S A_S\}^{(i)} - \{V_S A_S\}^{(i-1)} = \frac{1}{6} v_S \left(A_S^{(i-2)} - 6A_S^{(i-1)} + 3A_S^{(i)} + 2A_S^{(i+1)} \right)$$

The symmetrical scheme for the diffusion term is similar to that of A_{BR} , thus, for any $i \in \llbracket 2, n-1 \rrbracket$,

$$\begin{aligned}
D^{(i)} \frac{\partial}{\partial z} [A_S^{(i)}] - D^{(i-1)} \frac{\partial}{\partial z} [A_S^{(i-1)}] &= \frac{1}{\Delta z} \left(D^{(i)} (A_S^{(i+1)} - A_S^{(i)}) - D^{(i-1)} (A_S^{(i)} - A_S^{(i-1)}) \right) \\
&= \frac{1}{\Delta z} \left(D^{(i-1)} A_S^{(i-1)} - (D^{(i-1)} + D^{(i)}) A_S^{(i)} + D^{(i)} A_S^{(i+1)} \right)
\end{aligned}$$

Therefore, for any $i \in \llbracket 3, n-1 \rrbracket$,

$$\begin{aligned}\frac{J_{A_S}^{(i)} - J_{A_S}^{(i-1)}}{\Delta z} &= \frac{v_S}{6\Delta z} \left(A_S^{(i-2)} - 6A_S^{(i-1)} + 3A_S^{(i)} + 2A_S^{(i+1)} \right) \\ &\quad - \frac{1}{\Delta z^2} \left(D^{(i-1)} A_S^{(i-1)} - (D^{(i-1)} + D^{(i)}) A_S^{(i)} + D^{(i)} A_S^{(i+1)} \right)\end{aligned}$$

At the surface of the water column, the boundary condition imposes $J_{A_S}^{(0)} = 0$, and it holds

$$\{V_S A_S\}^{(1)} = \frac{1}{2} v_S (A_S^{(1)} + A_S^{(2)})$$

hence, for $i = 1$,

$$\frac{J_{A_S}^{(1)} - J_{A_S}^{(0)}}{\Delta z} = \frac{1}{2\Delta z} v_S (A_S^{(1)} + A_S^{(2)}) - \frac{1}{\Delta z^2} D^{(1)} (A_S^{(2)} - A_S^{(1)})$$

and, for $i = 2$,

$$\frac{J_{A_S}^{(2)} - J_{A_S}^{(1)}}{\Delta z} = \frac{1}{6\Delta z} v_S (-4A_S^{(1)} + 2A_S^{(2)} + 2A_S^{(3)}) - \frac{1}{\Delta z^2} (D^{(1)} A_S^{(1)} - (D^{(1)} + D^{(2)}) A_S^{(2)} + D^{(2)} A_S^{(3)})$$

At the bottom of the water column, the boundary condition imposes $\frac{\partial}{\partial z} [A_S^{(n)}] = 0$ thus $A_S^{(n+1)} = A_S^{(n)}$,

then

$$\{V_S A_S\}^{(n)} = \frac{1}{6} v_S (-A_S^{(n-1)} + 7A_S^{(n)})$$

hence, for $i = n$,

$$\frac{J_{A_S}^{(n)} - J_{A_S}^{(n-1)}}{\Delta z} = \frac{1}{6\Delta z} v_S (5A_S^{(n)} - 6A_S^{(n-1)} + A_S^{(n-2)}) - \frac{1}{\Delta z^2} D^{(n-1)} (A_S^{(n-1)} - A_S^{(n)})$$

S6.4. Computation of the nutrient flux derivative $\frac{\partial J_N}{\partial z} \Big|_{z=u_i, t}$

For all $i \in \llbracket 1, n \rrbracket$, it holds

$$\frac{\partial J_N}{\partial z} \Big|_{z=u_i, t} = \frac{J_N^{(i)} - J_N^{(i-1)}}{\Delta z} = \frac{1}{\Delta z} \left(-D^{(i)} \frac{\partial}{\partial z} [N^{(i)}] + D^{(i-1)} \frac{\partial}{\partial z} [N^{(i-1)}] \right)$$

Using the symmetrical scheme for the diffusion term, we have, for any $i \in \llbracket 2, n-1 \rrbracket$,

$$\frac{J_N^{(i)} - J_N^{(i-1)}}{\Delta z} = \frac{1}{\Delta z^2} (-D^{(i)} N^{(i+1)} + (D^{(i-1)} + D^{(i)}) N^{(i)} - D^{(i-1)} N^{(i-1)})$$

At the surface of the water column, the boundary condition imposes $J_N^{(0)} = 0$ thus, for $i = 1$,

$$\frac{J_N^{(i)} - J_N^{(i-1)}}{\Delta z} = -\frac{1}{\Delta z^2} D^{(1)} (N^{(2)} - N^{(1)})$$

At the bottom of the water column, the boundary condition imposes $\frac{\partial}{\partial z} [N^{(n)}] = \xi (N_0 - N^{(n)})$ thus, for $i = n$,

$$\begin{aligned} \frac{J_N^{(n)} - J_N^{(n-1)}}{\Delta z} &= \frac{1}{\Delta z} \left(-D^{(n)} \frac{\partial}{\partial z} [N^{(n)}] + D^{(n-1)} \frac{\partial}{\partial z} [N^{(n-1)}] \right) \\ &= -\frac{1}{\Delta z} D^{(n)} \xi (N_0 - N^{(n)}) + \frac{1}{\Delta z^2} D^{(n-1)} (N^{(n)} - N^{(n-1)}) \end{aligned}$$

S6.5. Computation of the light profile I

To compute the light profile, the integral term is computed using the trapezoidal rule (FERZIGER and PERIC, 1999; HUISMAN and SOMMEIJER, 2002) over the spatial grid. Thus, for $i = 1$,

$$I^{(1)} = I(u_1, t) = I_0 \exp \left(-a_{\text{BR}} \left(\frac{5}{8} A_{\text{BR}}^{(1)} - \frac{1}{8} A_{\text{BR}}^{(2)} \right) \Delta z - a_{\text{S}} \left(\frac{5}{8} A_{\text{S}}^{(1)} - \frac{1}{8} A_{\text{S}}^{(2)} \right) \Delta z - a_{\text{bg}} u_1 \right)$$

for $i = 2$,

$$I^{(2)} = I(u_2, t) = I_0 \exp \left(-a_{\text{BR}} \left(\frac{9}{8} A_{\text{BR}}^{(1)} + \frac{3}{8} A_{\text{BR}}^{(2)} \right) \Delta z - a_{\text{S}} \left(\frac{9}{8} A_{\text{S}}^{(1)} + \frac{3}{8} A_{\text{S}}^{(2)} \right) \Delta z - a_{\text{bg}} u_2 \right)$$

and, for any $i \geq 3$,

$$\begin{aligned} I^{(i)} = I(u_i, t) &= I_0 \exp \left(-a_{\text{BR}} \left(\frac{9}{8} A_{\text{BR}}^{(1)} + \frac{7}{8} A_{\text{BR}}^{(2)} + \sum_{k=3}^{i-1} A_{\text{BR}}^{(k)} + \frac{1}{2} A_{\text{BR}}^{(i)} \right) \Delta z \right. \\ &\quad \left. - a_{\text{S}} \left(\frac{9}{8} A_{\text{S}}^{(1)} + \frac{7}{8} A_{\text{S}}^{(2)} + \sum_{k=3}^{i-1} A_{\text{S}}^{(k)} + \frac{1}{2} A_{\text{S}}^{(i)} \right) \Delta z - a_{\text{bg}} u_i \right) \end{aligned}$$

Supplementary references

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